Sociodynamics of Emutopia and Kiwiland relations

Decoy saturation: when do cheap decoys break an air defence?

Story: Emutopia keeps sending “airborne humanitarian food deliveries” to Kiwiland, which would rather not receive them

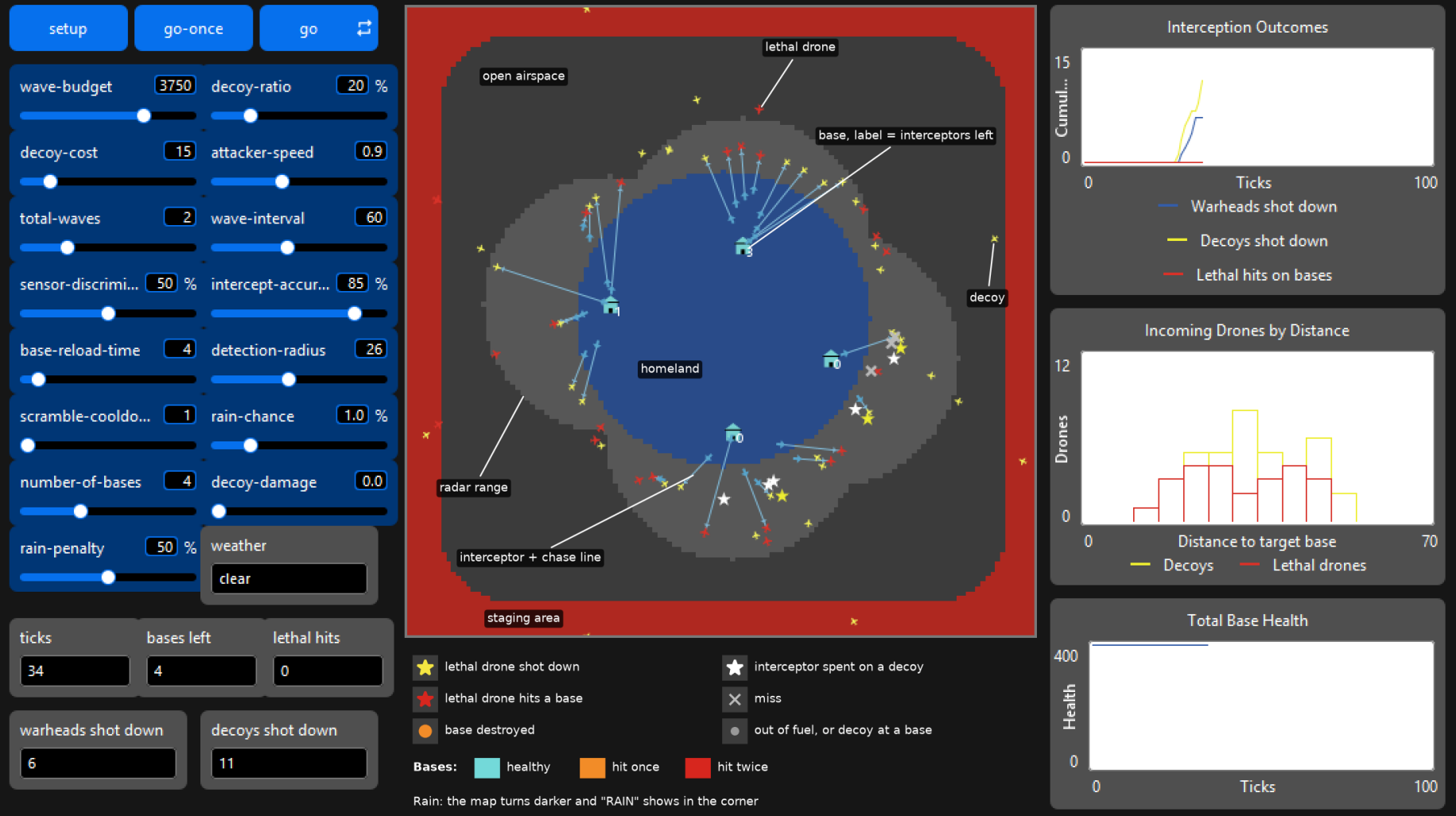
Question: how do the decoy share and decoy recognition decide if the defence holds?

Hypothesis: above some decoy share the magazines run dry and losses jump

Note: from here on we use military terms, as the situation is surprisingly close to a scenario of that type.

Matyas Patacsi (YA1O1L) · Collective Intelligence, Assignment 1

Model



Breeds

Bases 10 interceptors, slow reload

Drones lethal (red) or decoy (yellow)

Interceptors chase one drone, 85% kill

Rules

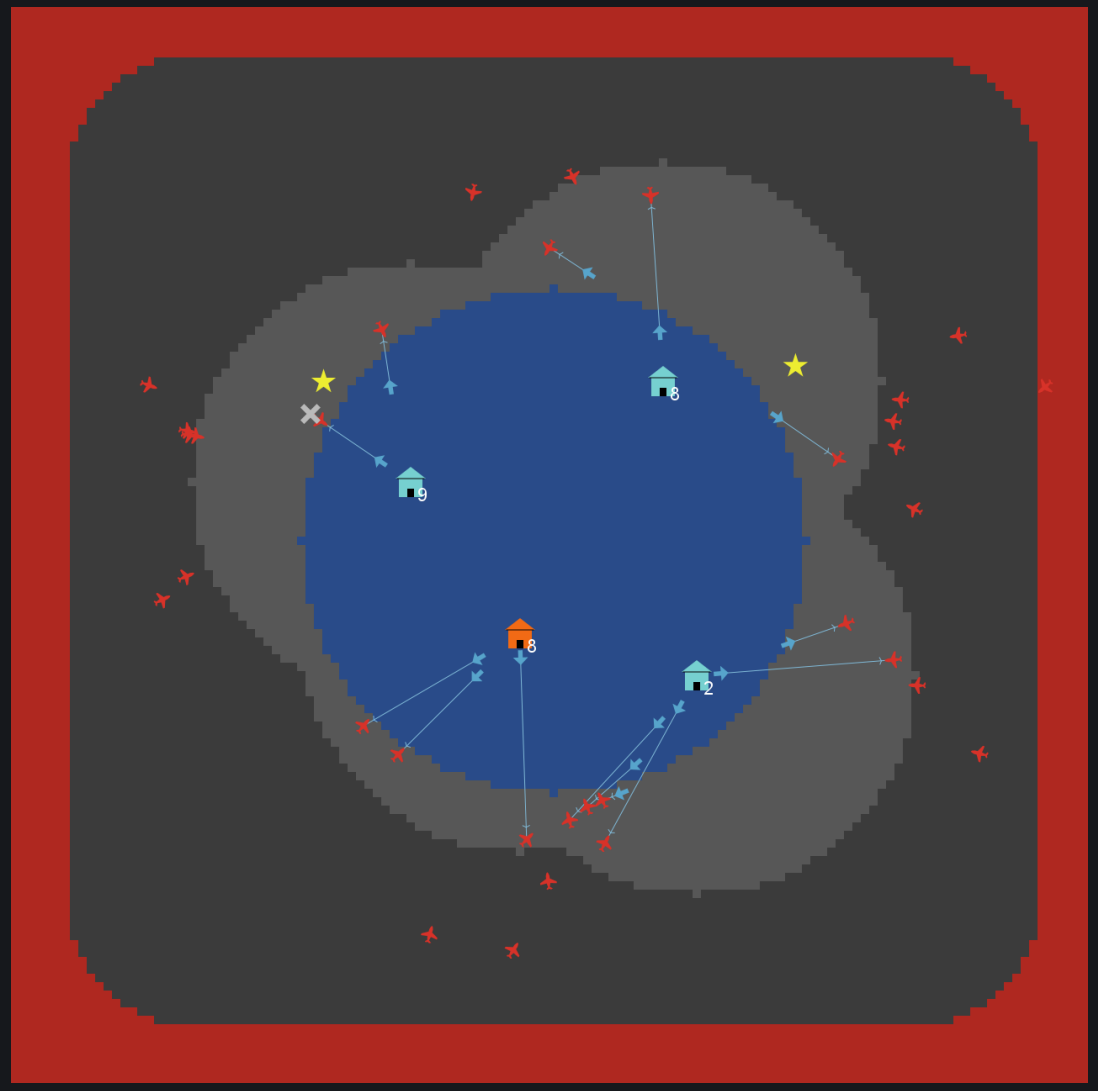
Decoy recognition aim at a lethal drone, or the nearest

Saturation no ammo, no launch

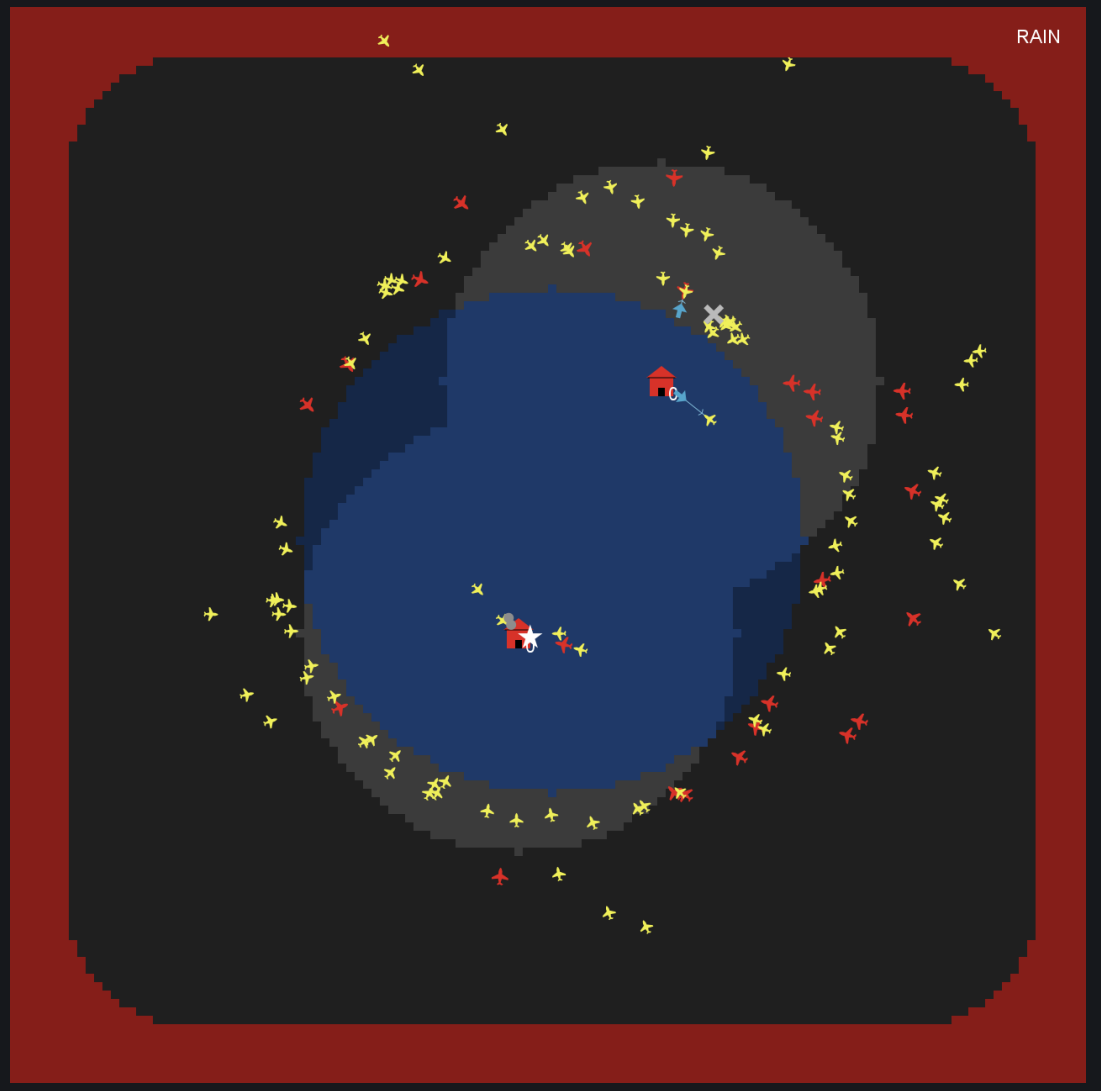
Rain halves the kill chance

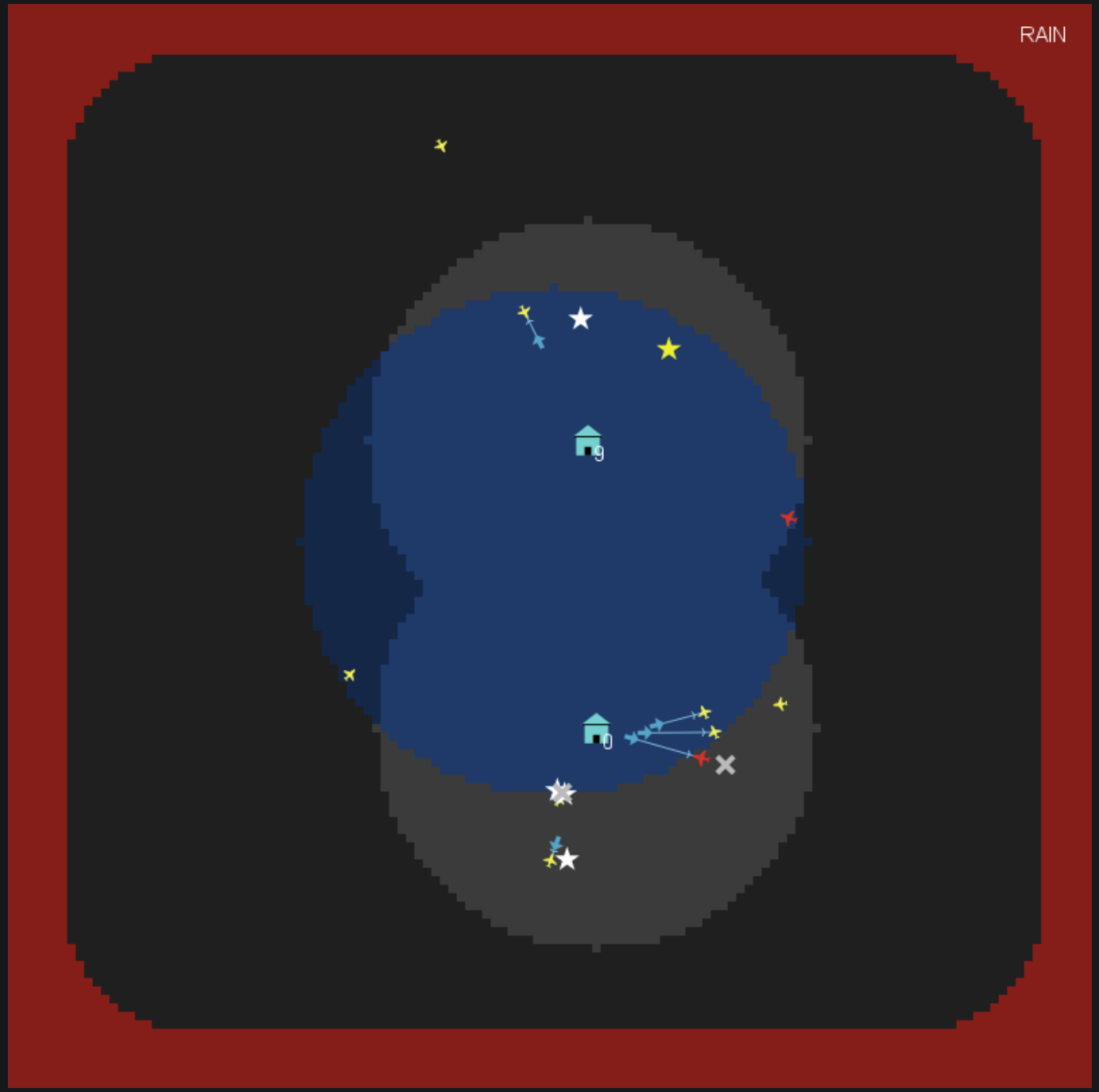
Simulation Snapshots

0% decoys



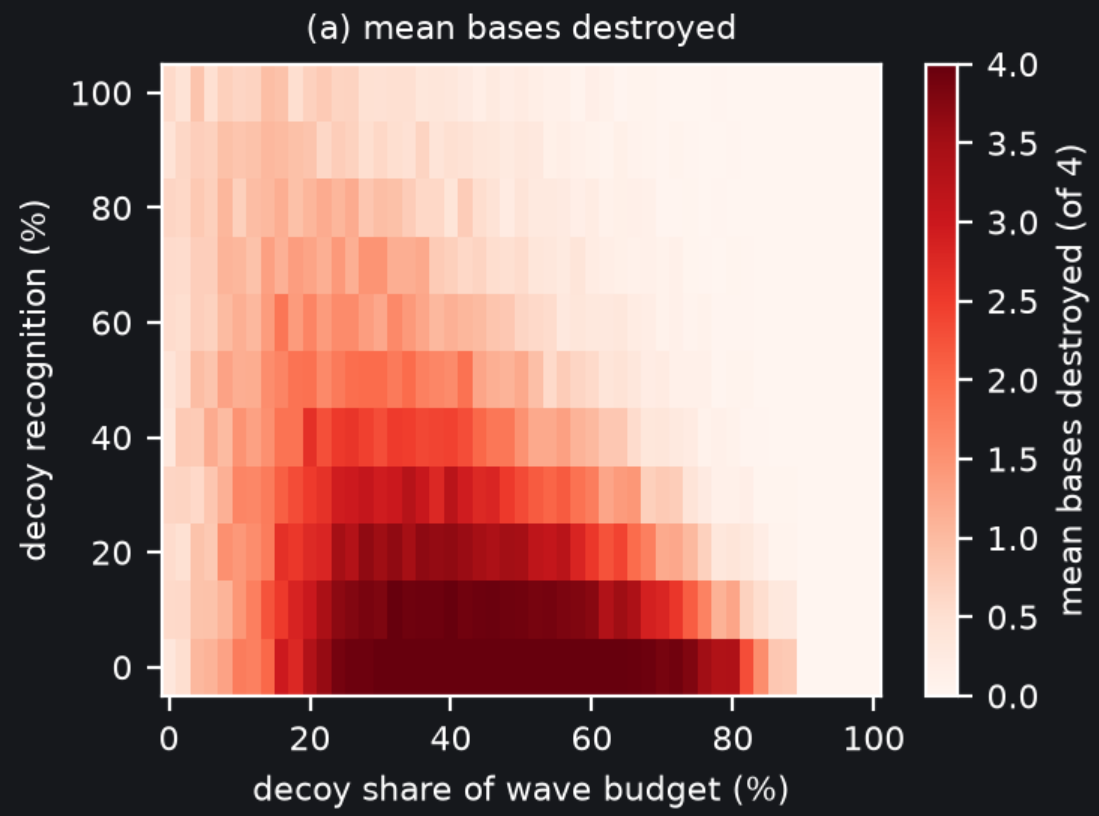
36% decoys

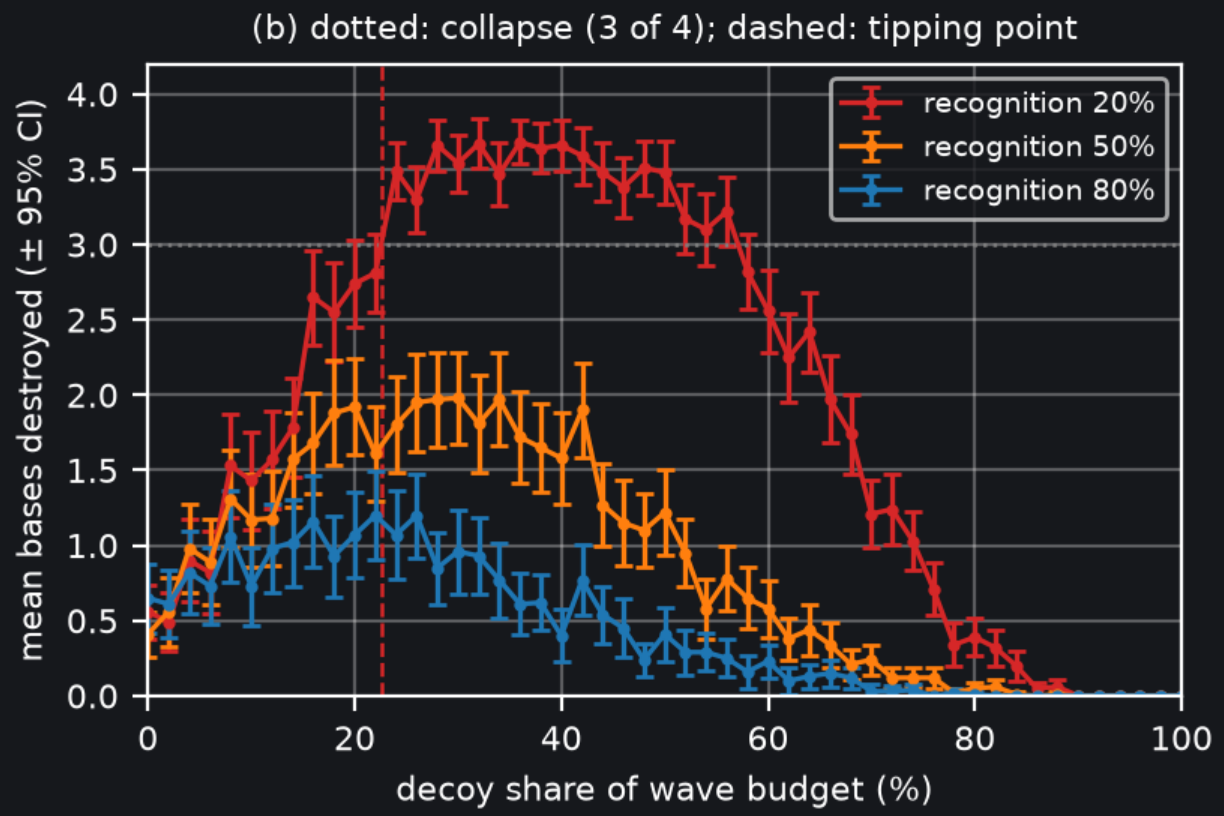



RAIN
9
0

Experiment: decoy share × decoy recognition

BehaviorSpace · 561 settings × 100 runs · bases destroyed, mean ± 95% CI

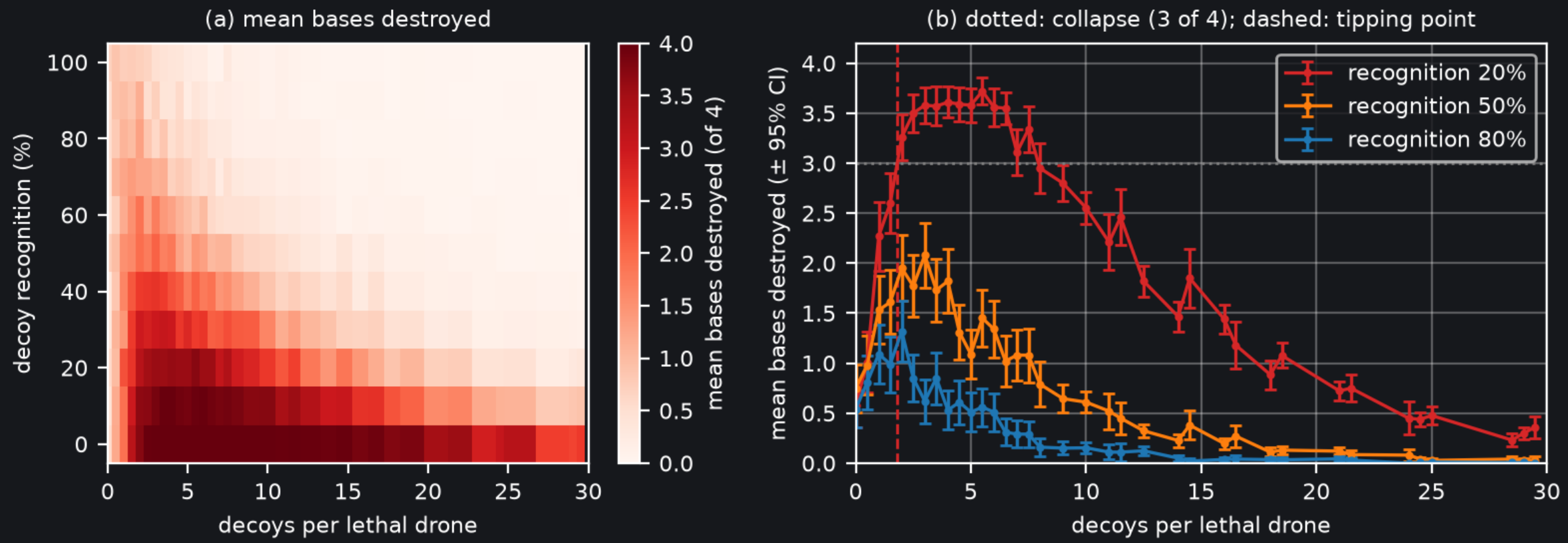




Tipping point: 22.6% decoys at 20% recognition

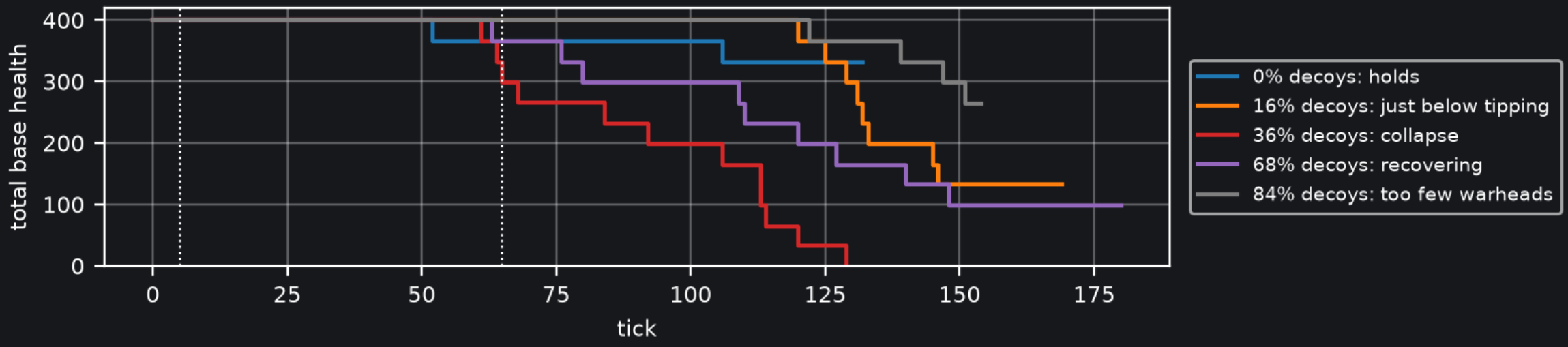
Experiment: decoy share × decoy recognition

BehaviorSpace · 671 settings × 100 runs · bases destroyed, mean ± 95% CI



Tipping point: about 1.8 decoys per armed drone at 20% recognition

Conclusions



Takeaway: the defense does not degrade slowly, it collapses once ~20% of the budget goes to decoys; after a certain point, there are not enough lethal drones for collapsing the defence

Limitations: straight-flying drones, simple fixed tactics on both sides, no adaptation or dynamic resource management

Thank you for your attention!

Questions?